DWS101 Μηχανική Μάθηση

Βιβλιογραφική Εργασία - Πρόβλεψη Ζήτησης Προϊόντων σε Καταστήματα Λιανικής

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Πίνακας περιεχομένων

1 Retail Forecasting Literature Review

Η εργασία αφορά στη μελέτη της βιβλιογραφίας και στην ανάπτυξη σύντομης παρουσίασης σχετικά με τη Πρόβλεψη Ζήτησης Προϊόντων σε Καταστήματα Λιανικής.

1.1 Οδηγίες

Η παρουσίαση θα πρέπει να περιλαμβάνει στο τέλος λίστα βιβλιογραφίας με τα άρθρα ή ηλεκτρονικές πηγές στις οποίες στηρίζεται. Την παρουσίαση ως αρχείο pdf/ppt και το video καταγραφής της ως mp4 (max 10-12 λεπτά) θα τα υποβάλετε ο καθένας σε ξεχωριστό φάκελο που θα έχει το όνομα σας και το ΑΕΜ σας, στο ακόλουθο link https://drive.google.com/drive/folders/1psePivGOgiiwfZ9DIC5rgtxHUKoZUnJj.

1.2 Χρήσιμοι Σύνδεσμοι

https://doi.org/10.1016/j.ijforecast.2021.11.013 https://doi.org/10.1007/s12351-020-00605-2 https://doi.org/10.1016/j.ijforecast.2021.11.001 https://doi.org/10.1016/j.ijforecast.2021.09.012

1.3 Εισαγωγή

Η πρόβλεψη πωλήσεων χρησιμοποιείται τα καταστήματα για τον προγραμματισμό του ανεφοδιασμού τους (επίπεδο καταστήματος) και της παραγωγής τους (επίπεδο αποθήκης). Πρόκειται για πρόβλεψη πολλών χρονοσειρών (όσα είναι τα προϊόντα) που μπορεί να αλληλεπιδρούν μεταξύ τους (π.χ. σε προϊόντα σουπερμάρκετ, ο πελάτης θα αγοράσει είτε γάλα πλήρες μαρκας Χ είτε μάρκας Υ), επηρεάζονται από ποικιλία προσφορών (μείωση τιμής, εκπτώσεις, 1+1, κλπ).

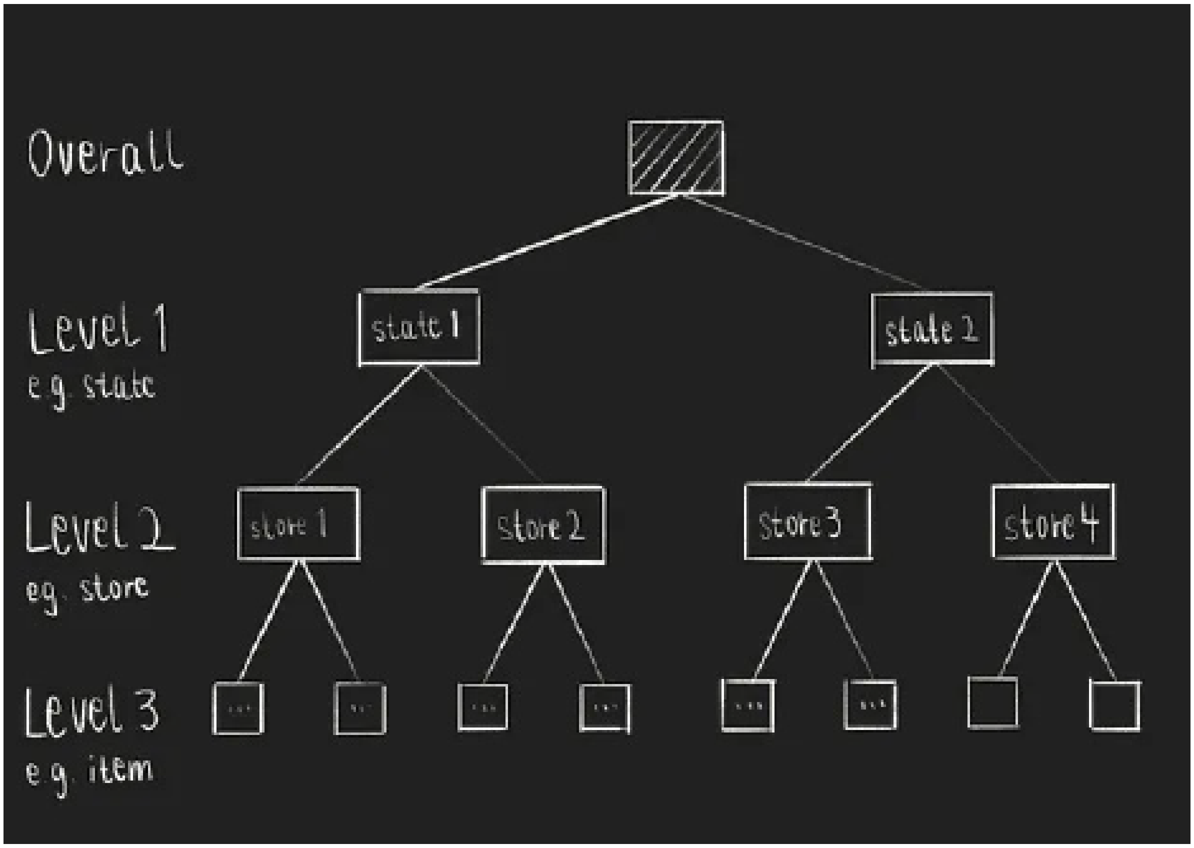
Forecasts are required in Super Markets in order to define:

- Strategy
- Manage demand, to avoid customer service issues, lost sales and low profits
- Manage stock supply, to reduce inventory costs

The data comprises aligned, highly-correlated series structured in a hierarchical fashion. complex, nonlinear relationships between series, as well as including exogenous/explanatory variables.

1.4 Όροι

Hierarchical unit sales refers to sales data structured in a multi-level grouping system, often used for forecasting, planning, reporting, and organizing sales teams. Unlike common multivariate time series problems, hierarchical time series can be aggregated on different levels: e.g., item level, store level, and state level.



Σχήμα 1: Hierarchical unit sales

Light Gradient-Boosting Machine (LGBM) is a free and open-source distributed gradient-boosting framework for machine learning, originally developed by Microsoft. It is based on decision tree algorithms and used for ranking, classification and other machine learning tasks. It is commonly adopted by retail firms to improve the accuracy of their sales predictions and daily operations.

Exponential Moving Average (or Exponential Smoothing) is an Optimizer, i.e. an algorithm or method used to minimize an error function (loss function). It gives exponentially decreasing weights to past observations, making *recent data more influential* for predicting the future than older data, unlike simple moving averages which weigh all points equally. It hence captures seasonality.

Exponential smoothing is an “off-the shelf” standard forecasting method that is readily available in most types of statistical software for anyone to use.

The Root Mean Squared Scaled Error (RMSSE) is a variant of the mean absolute scaled error (MASE), where we square and then take the root of errors. Scaling means taking into account the difference between each time point, by dividing by their mean. Normalizing means adjusting each error by dividing it by the observed sales.

One-step naive forecast means predicting the next period’s value will be exactly the same as the most recent actual value, δηλαδή θεωρώ την διαφορά τους ως «error», και αυτό βάζω στον παρονομαστή του RMSSE.

$$RMSSE = \sqrt{\frac{\frac{1}{h}\sum_{t=n+1}^{n+h}(y_t - \hat{y}_t)^2}{\frac{1}{n-1}\sum_{t=2}^{n}(y_t - y_{t-1})^2}}$$

- It is symmetric in the sense that it penalizes equally positive and negative forecast errors.
- It is therefore important to keep the values in a timeseries prior to the start of the selling of a product as na rather than 0, or RMSSE will calculated for them, too.

Global models: models that learn from the entire dataset to understand average behavior.

Naive forecast is Future sales = most recent observed sales, δηλαδή:

$$\hat{y}_{t+h} = y_t$$

όπου h: forecast horizon (in days)

Generalized Linear Models (GLM) are flexible statistical frameworks that extend traditional linear regression to handle response variables (outcomes) that aren’t normally distributed.

Autoregressive (AR) models predict future values in a sequence by using a linear regression of past values (lags) from the same variable, essentially modeling a variable's history to forecast its future behavior.

Recurrent Neural Networks: The output of a neuron at one time step is fed back as input to the network at the next time step. This enables RNNs to capture temporal dependencies and patterns within sequences.

recursive = recurrent = επαναλαμβανόμενος. Παίρνει προηγούμενα outputs ως inputs του επόμενου layer.

autoregressive = παίρνει προηγούμενες τιμές για να προβλέψει την επόμενη.

Mean shows accuracy and **standard deviation** shows stability.

Kurtosis is a measure of the risk of normally remote possibilities actually occurring. Ιτ happens when the probability of extreme outcomes rises, causing the tails on the bell curve to be elevated or "fattened." It measures the aggregate probability that outcomes in the two tails will occur: the greater the kurtosis, the more likely it is that extreme outcomes will come to pass.

Σχήμα 2: Kurtosis

Cross-learning = Global modelling, pooled learning. It's applicable in this case because we have a) the same nature of time series (retail) b) the same frequency c) hierarchical structure d) the same period of examination.

ARMA: The autoregressive (AR) component specifies that the current value of the series depends linearly on its own past values (lags), while the moving average (MA) component specifies that the current value depends on a linear combination of past error terms. ARMA works with stable (stationary) data, while ARIMA adds an "Integrated" (I) component, using differencing to make non-stationary data (with trends) stable before applying ARMA.

latent state: an underlying, unobservable condition or variable that influences observable data but isn't directly measured.

covariates: συναλλοιωτές, variables that influence the target in a study but aren't the main focus.

1.5 Traditional Practices

One decomposes the overall forecasting problem into a number of distinct forecasting sub-problems and applies a dedicated model or even a chain of models to each one. However, the forecasting problem might not lend itself to a decomposition into a sequence of distinct procedures, in which case one would have to think about model consolidation, blending, or ensembling. These techniques increase the complexity still further.

1.6 Methodology

One can simply forecast the series at the most disaggregated level of the competition and derive the remaining forecasts using a bottom-up approach. Alternatively, one might only forecast the most aggregated series (level 1) and compute the remainder using proportions (top-down method). A mixture of these two approaches is also possible (middle-out method).

1.7 Error Assessment

Regarding assessing the forecasting accuracy, those based on *scaled errors* probably have the most appropriate statistical properties.

In general, performing methods deteriorate at a lower aggregation level because the uncertainty increases when forecasting more disaggregated data with volatile sales, and patterns such as trend and seasonality are difficult to capture. In the lowest aggregation levels (10, 11, and

12) accuracy deteriorates significantly as the forecasting horizon increases, too. Longer-term trends are easier to observe at higher aggregation levels.

The weights for the WRMSSE are computed based on the cumulative actual dollar sales in each series (i.e. each product) in that particular period - hence offer more value to the companies by accounting more for higher-selling products.

We need to consider the full distributions of forecasting errors and especially their tails when evaluating forecasting methods, thereby indicating that robustness is a prerequisite for achieving high accuracy. That should be accounted for when cross-validating.

1.8 Exogenous factors

$$y_t = f(historyofy) + g(exogenousfactors_t) + \varepsilon_t$$

- is_christmas
- is_black_friday
- promo_active
- store_closed

However, Walmart's operating costs were not factored in.

1.9 Forecasting methods of the winning teams

Many of these sequences include intermittent sales having many zeros as well as high-volume sales. Intermittent data forecasting is concerned with sequences in which values appear sporadically.

1. Equal weighted combination (arithmetic mean) of 3 LightGBM models at the level of Store, Department and Category, with 2 variations (recursive & non-recursive) for each, adding up to 6 for each time series.

- Pooled per store: The model learns patterns that are shared across items of a specific store, for example sales on Sundays, or holidays and weather effects.
- **Recursive ('επαναληπτικό') Forecasting:** A *recursive approach* generates multi-step forecasts by feeding previous predictions back into the model, whereas a *non-recursive approach* predicts each future horizon step directly without using its own forecasts as inputs.
- Early stopping is a technique used during the training of machine learning models to prevent overfitting. Instead of training for a fixed number of epochs, early stopping monitors the model's performance on a validation set and halts training when performance begins to degrade.
- Υπέθεσε πως οι χρονοσειρές ακολουθούν κατανομή Tweedie. Η κατανομή Tweedie για $p > 1$ γίνεται εκθετική, δηλαδή η συνάρτηση μάζας - πιθανότητάς της ανήκει στην Exponential Dispersion Family.
- i.i.d.: independent and identically distributed (προέρχονται από τον ίδιο πληθυσμό) variables = random sample.

According to his validations, the recursive models in his approach were more accurate on average than the non-recursive models but with greater instability.

2. In this case, 10 LightGBM models were trained per store (pooled per store) and 5 N-BEATS multipliers were used to capture trends.

N-BEATS: a neural network that also predicts weekly sales and is especially good at capturing trends (i.e. upward/downward movement over time). In practice, we multiply LightGBM forecasts by some multiplier to bring them in line with N-BEATS trends. The reason is that the LightGBM model might be good at capturing seasonal patterns and short-term fluctuations (e.g. promotions), but not long-term trends. The multiplier adjusts the LightGBM outputs to match the trend predicted by N-BEATS. The concept can be summed as reconciling the forecasts produced at different aggregation levels.

Why use several multipliers

Each series of the product-store level (the lowest level) in the data set was forecast using a combination of five different models. Rather than using one multiplier for all weeks, one might choose to split the forecast horizon into segments (e.g., short-term, medium-term, long-term) and apply a different multiplier to each. This allows the adjusted forecasts to capture changing trends over time, rather than using a single static factor. For example, 5 multipliers could correspond to Weeks (1–2, 3–4, 5–6, 7–8, 9–10), where each segment gets its own multiplier derived from N-BEATS vs LightGBM predictions.

Loss function used: An asymmetric loss function penalizes overprediction and underprediction differently, unlike standard symmetric losses (like Mean Squared Error) that treat both errors equally. These functions are crucial in applications where one type of error is more costly.

No recursive (autoregressive) use of sales data in the LightGBM models, so there no dependence on previous forecasts to generate the next one. This helps to a) avoid forecast errors being propagated over the horizon and b) Long-range forecasts to remain stable.

3. This method used DeepAR, a specific type of recurrent neural network (RNNs) designed to model sequential data over time

For advanced time-series forecasting, Amazon Corporation developed a state-of-the-art probabilistic forecasting algorithm which is known as the Deep Autoregressive or DeepAR forecasting algorithm. This is one kind of Deep Learning model that is specifically designed to capture the inherent uncertainties associated with future predictions, with the ability to extract features from high-dimensional inputs. Unlike traditional forecasting methods that rely on deterministic point estimates, DeepAR provides a probability distribution over future values, allowing decision-makers to assess the range of possible outcomes and make more informed decisions.

Each neural network in the ensemble was built by stacking multiple Long Short Term Memory (LSTM) layers. The name of LSTMN reflects its key capabilities:

- Short-term memory: the ability to retain recent information from the sequence (e.g., the last few time steps).
- Long-term memory: the ability to preserve important information over long time spans and prevent it from being overwritten or forgotten.

24 of the models employed dropout, whereas 19 didn't. Dropout randomly deactivates neurons during training to reduce overfitting. However, if e.g. the dataset is large enough or other methods of overfitting (such as early stopping) are employed, it might not be needed.

CV here: last fourteen 28 day-long windows of available data (28 x 14 = 392 days).

Calibrating forecasts

Each LSTM is trained to output parameters of a probability distribution for future demand. So for *every future time step*, the model outputs a full predictive distribution, i.e. a mean parameter with possibly dispersion parameters. The NNs considered 100 features.

For each model i, horizon h, and series we have a Tweedie probability distribution $\tilde{y}_{i,h}$. We then draw S samples:

$$\tilde{y}_{i,h}^{(1)}, \dots, \tilde{y}_{i,h}^{(S)}$$

These samples represent possible future demand paths. We optimize the weights for the 43 NNs of the final model by training on these samples; so optimal and not equal combination weights are used in the final model.

Γιατί δημιουργώ τεχνητές μελλοντικές προβλέψεις;

The predictive distributions originate from the LSTM models themselves, which are trained to output full probability distributions for future demand. Θεωρούμε δηλαδή πως κάθε μοντέλο έχει κάνει ικανοποιητικές προβλέψεις (τις περάσαμε και από cross-validation), που κατ' εκτίμηση αποτυπώνουν τις πιθανές μελλοντικές τιμές. We then draw samples from those distributions με ίσες πιθανότητες για την τιμή που πρόβλεψε το κάθε μοντέλο (καθώς έχουμε ένα equal-weight ensemble) and create a target class for the 28 days. We then create a Tweedie General Linear Model where the inputs (features) are the 43 models, with weights applied to each, in order to get the targets we set. Το τελικό μας μοντέλο είναι δηλαδή ένα παραμετρικό μοντέλο, με τα χαρακτηριστικά του να είναι νευρωνικά μοντέλα, καθένα από τα οποία έχει 100 χαρακτηριστικά.

Είναι ένας τρόπος βελτιστοποίησης των μελλοντικών προβλέψεων καθώς: - Δεν υπάρχουν διαθέσιμες προβλέψεις για το μελλοντικό διάστημα των 28 ημερών, όμως οι προβλέψεις από το ensemble των μοντέλων καλύπτουν αυτό το διάστημα, με τα ιδιαίτερα χαρακτηριστικά του (π.χ. αργίες ή εποχικότητα). Έτσι μπορούμε να βελτιστοποιήσουμε τις προβλέψεις για το test. - Μπορούμε να εξασφαλίσουμε πως οι προβλέψεις θα έχουν την κατανομή που θέλουμε (π.χ. Tweedle), με τον αντίστοιχο αριθμό μηδενικών τιμών ή διακύμανσης. - Αποφεύγουμε την υπερπροσαρμογή σε θόρυβο των δεδομένων εκπαίδευσης.

Γιατί παίρνω δείγμα από ένα μοντέλο και όχι τον μέσο όρο τους;

Για να διατηρήσω την αβεβαιότητα και την τυχαιότητα στην εκπαίδευση του GLM. Διαφορετικά θα είχα «εξομαλυμένες» προβλέψεις, χωρίς ακραίες ή μηδενικές τιμές.

4. Εδώ έχουμε ένα non-recursive LGBM μοντέλο για κάθε Store (άρα 10 στο σύνολο). Το κάθε ένα από αυτά τα μοντέλα διαφοροποιήθηκε για τις 4 εβδομάδες των προβλέψεων, άρα συνολικά δημιουργήθηκαν 40 μοντέλα. Ως Cross Validation χρησιμοποιήθηκαν οι 5 τελευταίες περίοδοι 28 ημερών των δεδομένων εκπαίδευσης.

5. Σε αυτή την περίπτωση είχαμε εκπαίδευση μοντέλων ανά τμήμα καταστήματος (π.χ. τρόφιμα, είδη σπιτιού, ρούχα), άρα 7 μοντέλα. Οι μελλοντικές τιμές των προβλέψεων που λήφθησαν για κάθε προϊόν στη συνέχεια προσαρμόστηκαν ώστε ο μέσος όρος τους (των τιμών δηλαδή των 10 καταστημάτων) να είναι ίσος με τον μέσο όρο των δεδομένων εκπαίδευσης, για τις τελευταίες 28 ημέρες. Validation: Τυχαίο δείγμα 500 (!) ημερών.

link function in GLMs

In generalized linear models, there is a *link function*, which is the link between the *mean* of Y on the left and the fixed component on the right, μια συνάρτηση που συνδέει τον μέσο όρο της μεταβλητής στόχου με τα δεδομένα εισόδου.

$f(\mu_{Y|X}) = \beta_0 + \beta_1 X$

π.χ. $ln(\mu) = \beta_0 + \beta_1 X$ => $\mu = e^{(\beta_0 + \beta_1 X)}$

Γιατί χρησιμοποιώ την Maximum Log Likelihood;

Θέλω να βρω μια συνάρτηση που περιγράφει την κατανομή των δεδομένων μου ώστε:

- Να μπορώ να κάνω νέες προβλέψεις
- Να μπορώ να εκτιμήσω την επίδραση των χαρακτηριστικών
- Να μπορώ να υπολογίσω σφάλματα και διαστήματα εμπιστοσύνης

1.10 Conclusions

The results support the long-standing belief that combining forecasts obtained with different methods can improve the forecasting accuracy and they also confirm that there is no guarantee that an “optimal” forecast combination will perform better than a simpler, equal-weighted one.

Πλεονέκτημα έναντι άλλων μεθόδων:

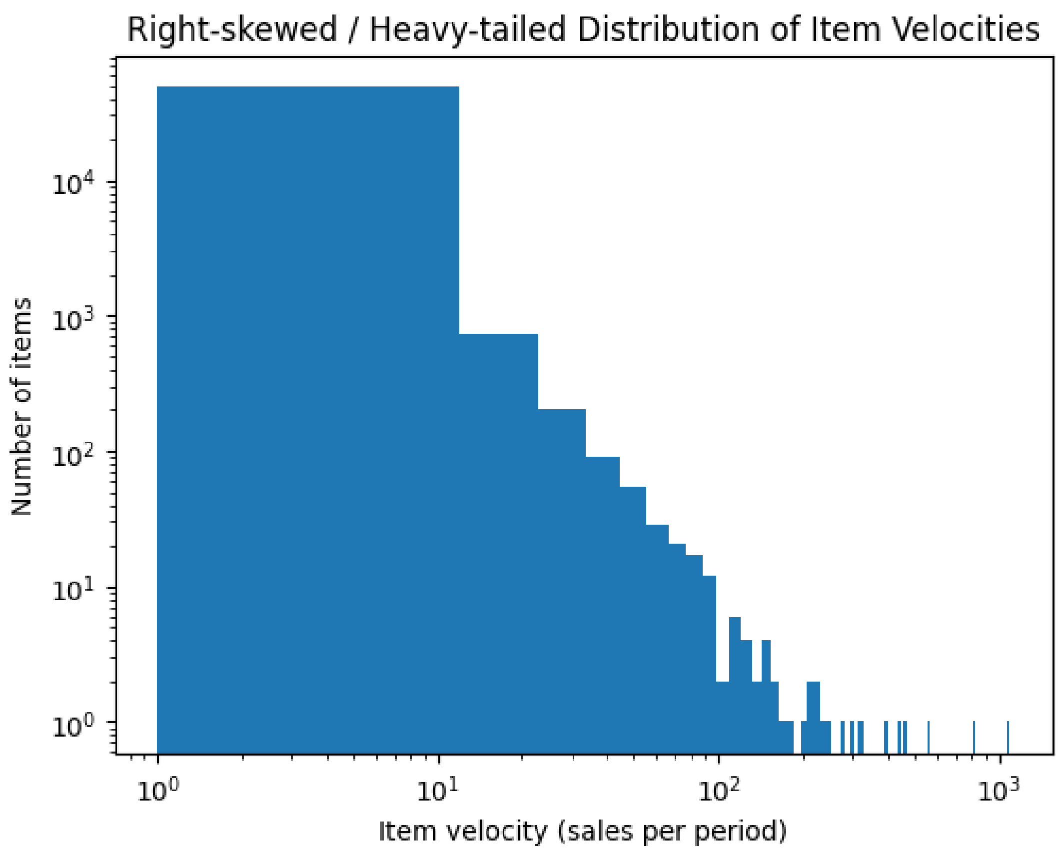
- Το να μην διαλέγεις χειρωνακτικά τα μοντέλα βάση αυτοσυσχέτισης, τάσεων και εξωγενών παραγόντων (avoid selecting and preparing covariates and selecting models that classical techniques require).
- Κάνει πιθανοτικές προβλέψεις αντί για σημειακές.

1.11 DeepAR

It’s an autoregressive, Recurrent Neural Network, which means it can model the sequential nature of time series data. It’s a global model, so it learns from the whole dataset to understand behaviour. RNNs are generally deterministic, in the sense that for the same x, $h_{(t-1)}$ and θ they produce the same output. However, we can make their output probabilistic by sampling from a likelihood distribution (e.g. Tweedie or Negative Binomial).

Issues to resolve:

- Different magnitude of sales per item (item velocity).
- Lack of homoscedasticity, stationarity and Gaussianity of data.



Σχήμα 3: Rightly-skewed valocity in sales

This makes binning or normalization very difficult.

The poisson distribution assumes that the mean ~ variance = λ. The Negative Binomial (NB) introduces an (extra) dispersion parameter that allows a greater variance and is used for count time series ($y \in \{1, 2, 3 \ldots\}$). The Tweedie distribution is used for sales time series, as it's continuous ($y \in \mathbb{R}$).

Normally, to capture complex behaviour of time series we would utilize a) manual covariates with one-hot encoding (e.g. is it holidays?) b) Fourier terms such as sin(2πkt/24) and cos(2πkt/24) for daily cycles, and sin(2πkt/168) and cos(2πkt/168) for weekly cycles, where (t) is time and (k) is the harmonic number (1, 2, 3…). The latter here is mostly obsolete.

The model produces probabilistic outputs using Monte Carlo simulation. That means it's run many times, each run produces a plausible future trajectory and the collection of these trajectories approximates the forecast distribution.

It also employs Long Short-Term Memory layers to alleviate the vanishing or exploding gradient problem. As an RNN, it also allows for encoding and decoding of different lengths.

RNNs

The transit (or transition) function ▯(▯) defines how the system moves (transits) from one state to the next.

h(t)=f(h(t−1),x(t);θ)

where:

- h(t−1): previous state of the system
- x(t): external input (or driving signal) at time t
- θ: parameters of the model

An RNN is defined by a recurrence of the form:

$$h_t = f(W_h h_{t-1} + W_x x_t + b)$$

- The same weight matrices (W_h, W_x) and bias (b) are used at every time step t.
- This shared parameter logic is what defines the recursive (or recurrent) structure.